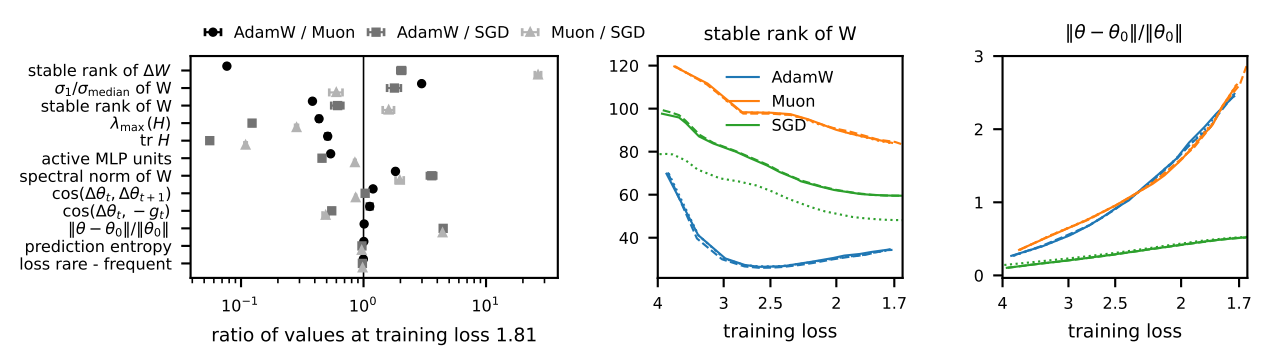
AdamW / Muon
AdamW / SGD
Muon / SGD
stable rank of ΔW
$\sigma_1/\sigma_{\text{median}}$ of W
stable rank of W
$\lambda_{\max}(H)$
tr H
active MLP units
spectral norm of W
$\cos(\Delta\theta_t, \Delta\theta_{t+1})$
$\cos(\Delta\theta_t, -g_t)$
$\|\theta-\theta_0\|/\|\theta_0\|$
prediction entropy
loss rare - frequent
10^{-1}
10^{0}
10^{1}
ratio of values at training loss 1.81
stable rank of W
AdamW
Muon
SGD
120
100
80
60
40
4
3
2.5
2
1.7
training loss
$\|\theta-\theta_0\|/\|\theta_0\|$
3
2
1
0
4
3
2.5
2
1.7
training loss